

Hanjia Lyu

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Research Areas

AI and Society, Multimodal LLMs, Graph Learning, Computational Social Science, Health Informatics

Education

Ph.D. in Computer Science, University of Rochester 2021 - 2026

Advisor: Prof. Jiebo Luo

M.S. in Data Science, University of Rochester 2019 - 2020

B.A. in Economics, Fudan University 2013 - 2017

Employment

Snap

Research Scientist Intern

Remote

May 2025–Apr 2026

Meta

Research Scientist Intern

Menlo Park, CA

Jun 2024–Jan 2025

Research Scientist Intern

Apr–Aug 2023

Research Scientist Intern

May–Sep 2022

Wyze Labs

Research Scientist Intern

Remote

Jan–May 2022

University of Rochester Medical Center

Research Assistant

Rochester, NY

Jan–May 2021

Recognition

◇ NSF Student Travel Award, ACM International Conference on Web Search and Data Mining (WSDM) 2026

◇ **Google PhD Fellowship** 2024

◇ Graduate Student Supplemental Professional Development Funding Award 2024

◇ NSF Student Travel Award, ACM International Conference on Information and Knowledge Management (CIKM) 2024

- ◇ Student Travel Award, IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR) 2024
- ◇ Student Travel Award, International Conference on Learning Representations (ICLR) 2024
- ◇ **Donald M. and Janet C. Barnard Fellowship** 2023
- ◇ **Snap Research Fellowship Semi-Finalist** 2022
- ◇ Graduate Student Scholarship, International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS) 2021
- ◇ First place in the UR Biomedical Data Science Hackathon 2020

Service to the Research Community

Area Chair

- ◇ Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) 2025
- ◇ Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024

Senior Program Committee

- ◇ International AAAI Conference on Web and Social Media (ICWSM) 2026-2027

Guest Editor for Journals

- ◇ Frontiers in Public Health
- ◇ ACM Transactions on Multimedia Computing, Communications, and Applications
 - Special Issue on Multimodal Video Understanding and Analysis with Foundation Models

Organizing Committee

- ◇ 1st Multimodal Learning for Social Good (MML4SG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024
- ◇ 1st Visual-Language Alignment in Text-Guided Multi-Modal Generation (TMMG) workshop at the IEEE International Conference on Multimedia and Expo (ICME) 2024

Selected Publications

[For a complete list of my publications, please visit [my homepage](#).]

1. **Hanjia Lyu**, Jiebo Luo, Jian Kang, and Allison Koenecke. Characterizing bias: Benchmarking large language models in simplified versus traditional Chinese. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '25, pp. 2815–2846, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400714825. doi: 10.1145/3715275.3732182. URL <https://doi.org/10.1145/3715275.3732182>

→ *Presented at: FAccT 2025, IC2S2 2025 (Parallel Talk)*

2. **Hanjia Lyu***, Jinfa Huang*, Daoan Zhang*, Yongsheng Yu*, Xinyi Mou*, Jinsheng Pan, Zhengyuan Yang, Zhongyu Wei, and Jiebo Luo. GPT-4V(ision) as a social media analysis engine. *ACM Trans. Intell. Syst. Technol.*, 16(3), April 2025. ISSN 2157-6904. doi: 10.1145/3709005. URL <https://doi.org/10.1145/3709005>

→ *Presented at: Synchronous Reading Club*

3. Hanqing Zeng*, **Hanjia Lyu***, Diyi Hu, Yinglong Xia, and Jiebo Luo. Mixture of weak & strong experts on graphs. In *The Twelfth International Conference on Learning Representations, ICLR*, Vienna, Austria, May, 2024. URL <https://openreview.net/pdf?id=wYvuY60SdD>

→ *Presented at: ICLR 2024*

4. **Hanjia Lyu**, Song Jiang, Hanqing Zeng, Yinglong Xia, Qifan Wang, Si Zhang, Ren Chen, Chris Leung, Jiajie Tang, and Jiebo Luo. LLM-Rec: Personalized recommendation via prompting large language models. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 583–612, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-naacl.39. URL <https://aclanthology.org/2024.findings-naacl.39/>

→ *Presented at: NAACL 2024, DEFIRST Reading Group at Mila x Vector Institute*

5. **Hanjia Lyu***, Jinsheng Pan*, Zichen Wang*, and Jiebo Luo. Computational assessment of hyperpartisanship in news titles. *Proceedings of the International AAAI Conference on Web and Social Media*, 18(1):999–1012, May 2024. doi: 10.1609/icwsm.v18i1.31368. URL <https://ojs.aaai.org/index.php/ICWSM/article/view/31368>

→ *Media coverage: UofR News Center, Study Finds, ScienceBlog, EurekAlert!*

→ *Presented at: ICWSM 2024*

6. Xinyi Mou, Zejun Li, **Hanjia Lyu**, Jiebo Luo, and Zhongyu Wei. Unifying local and global knowledge: Empowering large language models as political experts with knowledge graphs. In *Proceedings of the ACM Web Conference 2024*, WWW '24, pp. 2603–2614, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400701719. doi: 10.1145/3589334.3645616. URL <https://doi.org/10.1145/3589334.3645616>
7. **Hanjia Lyu**, Aarsal Imtiaz, Yufei Zhao, and Jiebo Luo. Human behavior in the time of COVID-19: Learning from big data. *Frontiers in Big Data*, 6:1099182, 2023. URL <https://doi.org/10.3389/fdata.2023.1099182>
8. **Hanjia Lyu** and Jiebo Luo. Understanding political polarization via jointly modeling users, connections and multimodal contents on heterogeneous graphs. In *Proceedings of the 30th ACM International Conference on Multimedia*, MM '22, pp. 4072–4082, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450392037. doi: 10.1145/3503161.3547898. URL <https://doi.org/10.1145/3503161.3547898>
 - *Presented at: MM 2022, Meta, Western New York Image and Signal Processing Workshop*
9. **Hanjia Lyu**, Junda Wang, Wei Wu, Viet Duong, Xiyang Zhang, Timothy D. Dye, and Jiebo Luo. Social media study of public opinions on potential COVID-19 vaccines: informing dissent, disparities, and dissemination. *Intelligent Medicine*, 2(1):1–12, 2022. ISSN 2667-1026. doi: <https://doi.org/10.1016/j.imed.2021.08.001>. URL <https://www.sciencedirect.com/science/article/pii/S266710262100036X>
 - *Media coverage: News Medical*
 - *Presented at: Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*
10. **Hanjia Lyu**, Ziheng Zheng, and Jiebo Luo. Misinformation versus facts: Understanding the influence of news regarding COVID-19 vaccines on vaccine uptake. *Health Data Science*, 2022. URL <https://doi.org/10.34133/2022/9858292>
 - *Media coverage: Medical Economics, News Medical, News Wise, Medical Xpress*
 - *Presented at: Great Lakes Science Boot Camp for Librarians, CSUG Research Panel*
11. **Hanjia Lyu**, Yangxin Fan, Ziyu Xiong, Mayya Komisarchik, and Jiebo Luo. Understanding public opinion toward the #StopAsianHate movement and the relation with racially motivated hate crimes in the US. *IEEE Transactions on Computational Social Systems*, 2021. URL <https://doi.org/10.1109/TCSS.2021.3136858>
 - *Media coverage: UofR News Center, Rochester Beacon, Futurity*

12. **Hanjia Lyu**, Long Chen, Yu Wang, and Jiebo Luo. Sense and sensibility: Characterizing social media users regarding the use of controversial terms for COVID-19. *IEEE Transactions on Big Data*, 7(6):952–960, 2020. URL <https://doi.org/10.1109/2FTBDATA.2020.2996401>

→ Media coverage: *IEEE Spectrum*, *Tech Xplore*, *UofR News Center*

→ Presented at: *Experience Rochester*, *CSUG Research Panel*

Teaching Experience

Guest Lecturer

University of Rochester Nov 2024
CSC 511: Large Language Models

University of Rochester Sep 2023
CSC 440: Data Mining

Teaching Assistant

University of Rochester
CSC 480: Computer Models & Limitations Spring 2023
CSC 486: Computational Complexity Spring 2023
CSC 440: Data Mining Fall 2022
CSC 446: Machine Learning Spring 2022
CSC 440: Data Mining Fall 2021
CSC 440: Data Mining Fall 2020

Talks & Publicity

- ◇ Presentation at the ACM International Conference on Web Search and Data Mining (WSDM) Feb 2026
- ◇ Presentation at the ACM Conference on Fairness, Accountability, and Transparency (FAccT) Jun 2025
- ◇ Invited talk at Meta Jun 2025
- ◇ Presentation at the ACM International Conference on Information and Knowledge Management (CIKM) Oct 2024
- ◇ Presentation at the IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR) Aug 2024

◇ Presentation at the International AAAI Conference on Web and Social Media (ICWSM)	Jun 2024
◇ Presentation at the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)	Jun 2024
◇ Presentation at the International Conference on Learning Representations (ICLR)	May 2024
◇ Invited talk at Data Science Snack Series: Internship Panel, University of Rochester	Feb 2024
◇ Invited talk at Synchronous Reading Club	Jan 2024
◇ Invited talk at Fudan University	Nov 2023
◇ Invited talk at DEFIRST Reading Group at Mila x Vector Institute	Oct 2023
◇ Presentation at the IEEE International Conference on Digital Health (ICDH)	Jul 2023
◇ Presentation at the AS&E Graduate Research Symposium, University of Rochester	Mar 2023
◇ Invited talk at the Western New York Image and Signal Processing Workshop	Nov 2022
◇ Invited talk at Meta	Aug 2022
◇ Invited talk at the Great Lakes Science Boot Camp for Librarians	Jun 2022
◇ Invited talk at Fudan-UC Center on Contemporary China, UCSD	May 2022
◇ Invited talk at InsurTech Innovation: UK & China as Hotspots of the Global World	Apr 2022
◇ Invited talk at CSUG Research Panel, University of Rochester	Oct 2021
◇ Presentation at the International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS)	Jul 2021
◇ Invited talk at Experience Rochester, University of Rochester	Sep 2020